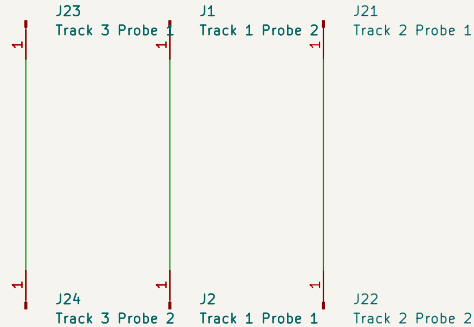


PCB Track Resistance



4-Wire Kelvin ACIR Measurement:

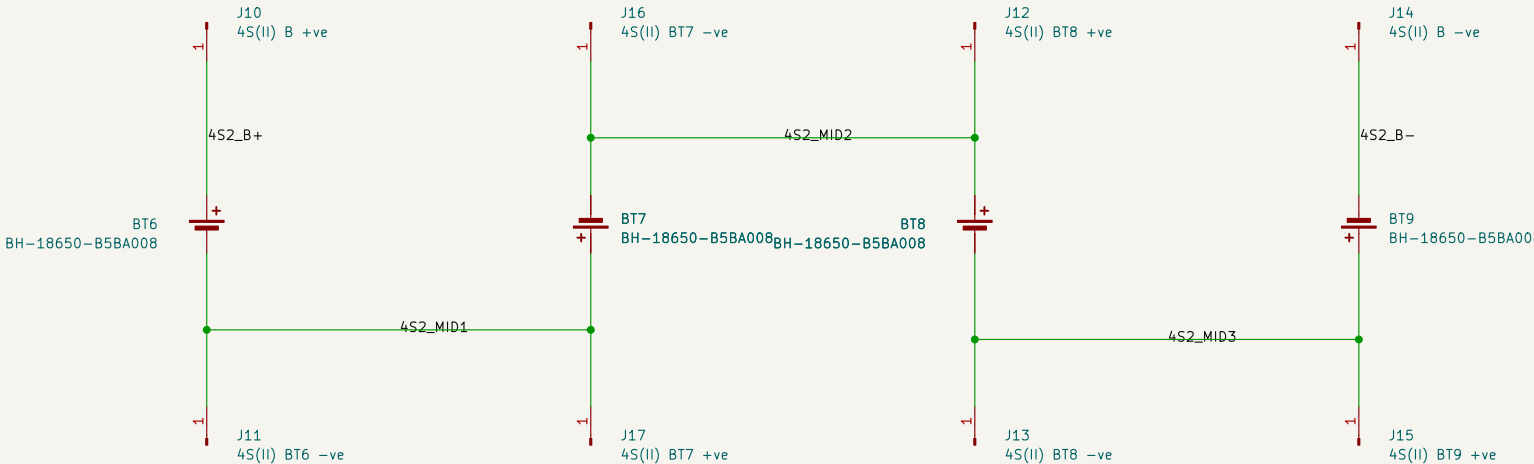
Inject 1 kHz AC current across B+ and B- terminals.

Sense AC voltage drop to isolate purely ohmic internal cell resistance.

4S 1P MODULE (Series String – Voltage Stacking & 4x ACIR)

4S 1P (II) MODULE (Varying Interconnect Track Resistance – Educational Pack Experiment)

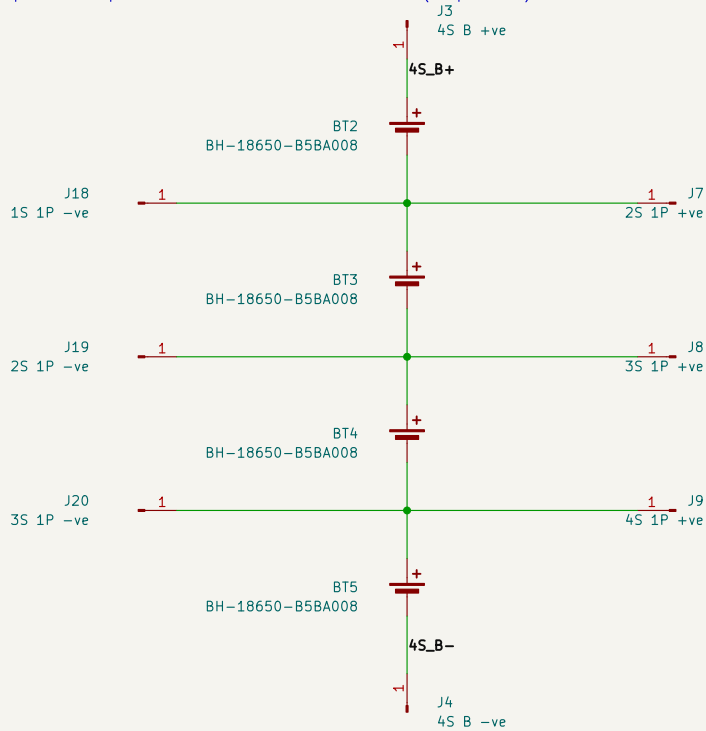
Current Distribution & Branch Kelvin Sensing: $1/R_{eq} = 1/R1 + 1/R2 + 1/R3 + 1/R4$ ($R_{eq} \approx R/4$)



Cells have IDENTICAL ACIR (~20mOhm); Series interconnects vary systematically:
Link 1 (BT6–BT7): Ultra–wide 10mm 2–layer bus with dense via matrix (~0.5mOhm)
Link 2 (BT7–BT8): Medium 3.0mm single–layer trace (~3.0mOhm)
Link 3 (BT8–BT9): Narrow 0.8mm single–layer trace (~10.0mOhm)
Demonstrates how interconnect trace impedance directly inflates total pack ACIR.

4S 1P MODULE (Series String – Voltage Stacking & 4x ACIR)

Total Series Impedance: $R_{eq} = R1 + R2 + R3 + R4 + R_{interconnect}$ ($R_{eq} \approx 4 * R$)



Inter–cell Kelvin probing pairs (J18/J7, J19/J8, J20/J9) allow measuring:

- Individual cell ACIR in–situ under series string current loading.
- Interconnect contact resistance across cell–to–cell busbars.

MECHANICAL MOUNTING HOLES

8x M3 Plated Chassis Grounding / Standoff Fixtures



EIS DESIGN & FABRICATION SPECS

1) Kelvin 4-Wire EIS Architecture:

Separate excitation and sensing paths eliminate test lead resistance.

2) Ultra-Dense Micro-Via Array Stitching:

1,792 stitched vias across high-current copper planes.

Effective parallel capacitance $C_{eff} < 0.001$ pF (negligible).

Halved DC resistance ($R_{dc} / 2$) for maximum measurement SNR.

3) PCB Stackup & Layer Specs:

2-Layer 1.6mm FR4, 1oz Cu, Matte Black Mask, HASL Lead-Free.

Production-Ready Hardware Design & Educational Measurement Kit High-Frequency 1kHz Kelvin 4-Wire EIS Sensing Architecture & Ultra-Dense Via Arrays Kit 2: PCB Track Resistance 4S1P (I) Differential ACIR 4S1P (II) Varying Interconnect IR Comprehensive 18650 ACIR Characterization Platform Decibels Lab Pvt Ltd		
Sheet: / File: ACIR-Learning-Kit-2-PCB.kicad_sch		
Title: ACIR LEARNING KIT 2 – CHARACTERIZATION & EXPERIMENTATION BOARD		
Size: A3	Date: 2026-09-16	Rev: v2.0
KiCad E.D.A. 10.0.5		Id: 1/1